The Transition to Electric Vehicles in the UK: Spatiotemporal Analysis and Future Trajectories

Group 1

Chenxi Hu, Jipeng Song, Yiding Wang, and Yuan Lu

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Note: This report fulfils the submission requirements as specified in the module guidelines.

Abstract

As the United Kingdom accelerates its transition toward a net-zero economy by 2050, the electrification of the transport sector remains a critical and highly complex challenge. Despite the aggressive regulatory framework established by the Zero Emission Vehicle (ZEV) Mandate, structural disparities in infrastructure and socio-economic wealth threaten to fragment adoption. This study presents a rigorous spatiotemporal analysis of electric vehicle (EV) penetration across the UK. By synthesizing multi-source data spanning over a decade, we identify severe spatial clustering and widening regional divides. Furthermore, we construct a mathematically bounded forecasting model, establishing an absolute fleet penetration ceiling of 50.24% by 2035. Through advanced exploratory data analysis and baseline machine learning evaluations, this paper highlights the urgent need for localised, targeted policy interventions to resolve infrastructural endogeneity and ensure an equitable national transition.

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1 Introduction

1.1 Research Background

Amid intensifying global efforts to mitigate anthropogenic climate change, the large-scale transition to renewable energy systems and zero-emission transport has emerged as both a geopolitical priority and an environmental necessity. The United Kingdom has enacted a statutory commitment to achieve net-zero greenhouse gas (GHG) emissions by 2050. Given that the transport sector constitutes a major source of national carbon emissions, its comprehensive and structural decarbonisation is a critical prerequisite for meeting this target. In this context, the Zero Emission Vehicle (ZEV) Mandate, introduced in 2024, marks a shift from policies primarily aimed at stimulating early adoption to a regime of mandatory sector-wide compliance. Under this framework, it is a legal requirement that 100% of new passenger car sales be zero-emission by 2035 Department for Transport (2023).

1.2 Problem Statement

Despite the substantial recent expansion of the battery electric vehicle (BEV) market, the pathway toward the planned 2035 transition remains impeded by significant systemic constraints. A key challenge is the growing disparity in charging and related infrastructure provision, manifested in pronounced spatial inequalities between affluent metropolitan centres such as London and more peripheral or rural areas. In addition, the interaction of elevated upfront purchase costs, range anxiety, and housing or property-related limitations (e.g., lack of off-street parking) generates a multifaceted and uneven adoption environment. Collectively, these barriers suggest that the national green transition is highly fragmented, thereby necessitating a fine-grained analysis of regional heterogeneity and its underlying socio-economic determinants.

1.3 Research Objectives

To address the aforementioned challenges, this report focuses on three primary research objectives:

- **Spatiotemporal Analysis of EV Adoption:** To explore the evolutionary patterns of electric vehicle (EV) penetration across the UK in recent years, identifying key geographical clusters and temporal trends using Exploratory Data Analysis (EDA).
- **Driver Analysis and Feature Weighting:** To extract and quantify the socio-economic and infrastructural drivers of EV uptake using machine learning frameworks (XGBoost and SHAP), mitigating the endogeneity of historical variables.
- **Predictive Modelling and Strategic Recommendations:** To develop a physically bounded predictive engine capable of forecasting regional EV penetration up to 2035, providing evidence-based recommendations for infrastructure planning.

2 Background Research

2.1 Emissions and Policy Context

The UK's decarbonisation strategy is anchored by its obligations under the Paris Agreement and the domestic Climate Change Act UNFCCC (2015); UK Government (2008). Currently, the transport sec-

tor represents the most significant hurdle, accounting for approximately 22% to 30% of total UK GHG emissions Climate Change Committee (2020), with passenger cars alone responsible for roughly 13%. Consequently, the Climate Change Committee (CCC) has identified the total electrification of the passenger fleet as the most critical milestone for achieving net zero Scottish Government (2018); Climate Change Committee (n.d.).

2.2 Progress and Structural Bottlenecks

To force this transition, the ZEV Mandate requires automakers to scale zero-emission sales to 33% by 2026 and 100% by 2035 Department for Transport (2023). However, infrastructural deployment is failing to keep pace with vehicle adoption. The national ratio of EVs to public charge points deteriorated significantly from 33:1 in 2019 to 95:1 in 2022 Wallace, Darwish, & Lai (2024). This creates a pervasive "chicken-and-egg" market failure: commercial operators delay network expansion due to low localised EV density Shi et al. (2021), while consumers resist EV adoption due to inadequate charging access. This failure is highly localised, with some regions suffering from a chargepoint deficit up to 3.2 times the national average, leaving rural and economically deprived areas at severe risk of being left behind Dong & Aston (2023).

The Sales vs. Fleet Discrepancy. A critical oversight in contemporary decarbonisation discourse is the conflation of "new-sales penetration" with "overall fleet penetration". While the ZEV Mandate legally enforces a 100% zero-emission new-car sales target by 2035, the UK passenger car fleet operates on a scrappage cycle of approximately 14 years. Consequently, even under perfect mandate compliance, millions of legacy internal combustion engine (ICE) vehicles will remain in active circulation long after the 2030 and 2035 milestones. Therefore, modelling the true ecological and infrastructural impact requires forecasting the absolute fleet stock rather than isolated sales flows. This structural reality establishes a strict mathematical ceiling on total fleet penetration, which this study rigorously incorporates into its predictive system dynamics.

Calculating the Absolute Ceiling (K) by 2035: To determine this ceiling, we constructed a Stock-Flow model based on official parameters:

1. *Legal Upper Limit:* The Vehicle Emissions Trading Schemes Order 2023 sets statutory yearly rates for ZEV new sales: 2024 (22%), 2025 (28%), 2026 (33%), 2027 (38%), 2028 (52%), 2029 (66%), 2030 (80%), incrementing linearly to 2035 (100%).

2. *Physical Base:* According to SMMT data, average annual new car sales in the UK remain stable at approximately 2,000,000, with a total existing car fleet of roughly 33,000,000.

Assuming perfect execution of the most optimistic government legislation starting from 2024, the total newly added EVs supplied to the road by the end of 2035 is modelled as:

$$\text{New EVs (2024-2035)} = 2,000,000 \times \sum_{t=2024}^{2035} \text{ZEV_Rate}_t$$

Summing the statutory proportions over these 12 years yields:

$$0.22 + 0.28 + 0.33 + \cdots + 1.0 = 7.79$$

$$2,000,000 \times 7.79 = 15,580,000 \text{ new EVs}$$

Given that the UK had an existing EV stock of approximately 1,000,000 at the end of 2023, the absolute penetration ceiling (K) is bound by:

$$\text{Absolute Ceiling } K = \frac{15,580,000 + 1,000,000}{33,000,000} = 50.24\%$$

This 50.24% ceiling represents the physical and mathematical limit of maximum fleet penetration by 2035.

2.3 Socio-Economic Determinants

Existing literature confirms that spatial variability in EV adoption is deterministic rather than random. For instance, Khader et al. demonstrated that spatial error models can explain 85.6% of the variance in adoption rates at the local authority district (LAD) level Khader, Wilson, & Tryfonas (2025). The transition is heavily gated by wealth and property types. High upfront capital costs naturally exclude lower-income households. Furthermore, approximately 34% of UK households lack off-street parking, rendering them entirely reliant on the fragmented public charging network. Conversely, households with private driveways—typically occupying higher social strata—enjoy significantly lower charging costs and operational convenience. Additionally, studies show that nearly 80% of early EV adopters own a secondary ICE vehicle, utilising the EV for urban commutes while relying on the ICE vehicle to bypass range anxiety on longer journeys Lee & Brown (2021).

2.4 Differentiation of this Study

While previous studies have successfully identified static correlations between income, infrastructure, and EV adoption, they predominantly rely on historical observation. This project differentiates itself by transitioning from static correlation to dynamic, policy-constrained forecasting. By integrating a SHAP attribution framework with a Panel Fractional Logit Model (PFLM), this study not only quantifies the non-linear tipping points of wealth and infrastructure but also projects how these constraints will geographically fracture the UK's 2035 ZEV targets if left unaddressed.

3 Question Development

The formulation of our research problem underwent a rigorous, iterative refinement process to ensure both academic viability and real-world policy relevance.

Stage 1: Broad Thematic Scoping. Initially, the project aimed to broadly analyse the UK's environmental strategies for carbon reduction. Given the overwhelming contribution of passenger cars to national GHG emissions and the aggressive legislative timeline of the ZEV Mandate, we quickly narrowed our focus to the electric vehicle transition.

Stage 2: Exploratory Pivot. Our initial assumption was to model national EV growth. However, early data exploration revealed that calculating a single UK-wide average masked severe micro-level realities. The geographical distribution of EVs and charge points was highly clustered, leading to a deteriorating infrastructure ratio in specific regions. This realisation forced a pivot: we determined that a macro-level study would be useless for practical policy. The problem needed to be addressed at the localised LAD level to uncover the socio-economic friction causing these disparities.

Stage 3: Methodological Refinement. Having identified disparities, we questioned why they existed. We realised that simply stating "wealthier areas have more EVs" was academically insufficient.

We needed to know the exact threshold at which wealth triggers adoption and the critical mass of infrastructure required to sustain it. This necessitated the use of machine learning (XGBoost + SHAP) to isolate non-linear driver weights. Finally, we recognized that explaining the past is only half the problem; to provide actionable advice, we needed to simulate the future.

4 Retrieving the Data

To comprehensively investigate the driving factors behind the adoption of electric vehicles (EVs) in the UK and their spatiotemporal evolution trends, this study constructs a multi-source, heterogeneous database that integrates infrastructure, transportation statistics, socioeconomic indicators, and geographical boundaries. The required data primarily cover the following four dimensions:

- **Infrastructure Data:** Used to measure the depth of coverage and temporal expansion of the charging network.
- **Vehicle Stock:** Used to calculate the penetration rate of electric vehicles and the evolution of energy structure.
- **Socioeconomic Indicators:** Including population size and Gross Disposable Household Income (GDHI), which serve as key variables for assessing market potential.
- **Macroeconomic Indicators and Geographical Boundaries:** Used for data interpolation (CPIH index) and spatial visualization (GeoJSON).

An overview of the specific datasets utilized is detailed in Table 4.1.

Table 4.1: Overview of Comprehensive Datasets Integrated for the Study

Dataset Name	Source Agency	Time Span	Key Variables
Electric vehicle public charging statistics	Dept for Transport (DfT)	2019 – 2025	Public charging devices, rapid/ultra-rapid counts
Vehicle Licensing Statistics: VEH0105/VEH0132	Dept for Transport (DfT)	2011 – 2024	Total licensed vehicles, ULEV/EV stock by region
Regional GDHI for Local Authorities	Office for National Statistics	2011 – 2022	GDHI per head, total household income
Detailed Time Series (England & Wales)	Office for National Statistics	2011 – 2024	Annual population estimates by age and gender
Mid-year population estimates (Scotland)	Nat. Records of Scotland	2011 – 2022	Annual population estimates by council area
Mid-year population estimates (NI)	NISRA	2011 – 2024	Annual population estimates by LGD
CPIH ANNUAL RATE 00: 2015=100	Office for National Statistics	2011 – 2025	Inflation indices for economic data imputation
UK-GeoJSON (2024 Edition)	Martinjc (GitHub / ONS)	2024	Digital vector boundaries (LAD, Region, Country)

5 Preparing the Data

To construct a robust spatiotemporal panel dataset, this study integrates multi-source, heterogeneous data from the aforementioned institutions. Given the significant differences in statistical standards, publication cycles, and spatial hierarchies among the raw data, we implemented a rigorous bottom-up data reconstruction and feature engineering process.

5.1 Spatial Hierarchy Mapping and Structural Processing

This study first converted all data sources into a longitudinal panel format, standardizing them into a "region-year-indicator" structure. During the multi-source data merging process, we developed a dynamic spatial mapping algorithm to address the complexities of the UK's administrative division system:

- **Spatial Nesting Mapping:** Using the ONS official geographic coding for infrastructure as a baseline, we constructed a complete nesting dictionary, mapping from Local Administrative Districts (LADs) to counties, regions, and finally, to the national level.
- **Symmetry Correction:** For Scotland and Northern Ireland, which lack a regional level division, the algorithm automatically downscales their national level data to ensure symmetry in the spatial matrix for the entire UK.
- **Population-Weighted Aggregation:** When aggregating absolute values (such as GDHI) to higher administrative levels, population-weighted methods were used to calculate the weighted per capita income for region R in year t, to avoid statistical distortion.

$$GDHI_{R,t} = \frac{\sum_{i \in R}(GDHI_{i,t} \times Pop_{i,t})}{\sum_{i \in R} Pop_{i,t}} \tag{5.1}$$

Here, i represents the lower-level micro LADs, and $Pop_{i,t}$ denotes the total population in year t.

5.2 Data Interpolation and Outlier Cleaning

Temporal Interpolation: Due to misalignment in publication cycles, an independent incremental population patch was developed. Missing population data for 2024 and 2025 were addressed using a time-series forward fill method, assuming short-term demographic stability. For economic data released with a 1–2 year lag, the Real Compound Annual Growth Rate (Real CAGR) and CPIH index were utilized to deflate nominal GDHI (stripping inflation) to calculate endogenous growth.

$$CAGR_i = \left(\frac{Real_GDHI_{i,2019}}{Real_GDHI_{i,2011}}\right)^{\frac{1}{8}} - 1 \tag{5.2}$$

Forecasted nominal values for $t \in \{2024, 2025\}$ were mapped via:

$$GDHI_{i,t} = GDHI_{i,t-1} \times (1 + CAGR_i) \times \left(1 + \frac{CPIH_Rate_t}{100}\right) \tag{5.3}$$

Variable Construction: Masked values ([x], [z], [low]) were replaced with nulls or zeros, and passenger vehicle stocks were adjusted to physical units. Three core metrics were derived:

1. *EV Penetration Rate:* Percentage of ULEVs in total passenger vehicle stock.
2. *Charging Infrastructure Density:* Ports per 100,000 population.
3. *EV-to-Port Ratio:* Ratio of EV stock to the number of charging ports.

To mitigate endogeneity (bidirectional causality) between infrastructure, wealth, and adoption, explanatory variables were treated as $T-1$ lagged variables. All continuous features underwent Z-score normalization. Geospatial aggregation algorithms further seamlessly merged Northern Ireland micro-polygons into macro-regions for UK-wide mapping.

6 Exploratory Data Analysis

Given the multi-source panel data containing extensive spatial and temporal information, our exploratory data analysis (EDA) extends beyond basic descriptive statistics. We developed a progressive exploration

strategy focusing on spatial distribution, temporal evolution, social dynamics, and economic logic.

6.1 Visual Exploration of Spatial Heterogeneity

Relying solely on macroscopic averages at the national level can obscure true variations at the local level. Therefore, this study integrated GeoJSON boundary data to construct multi-scale spatial heatmaps.

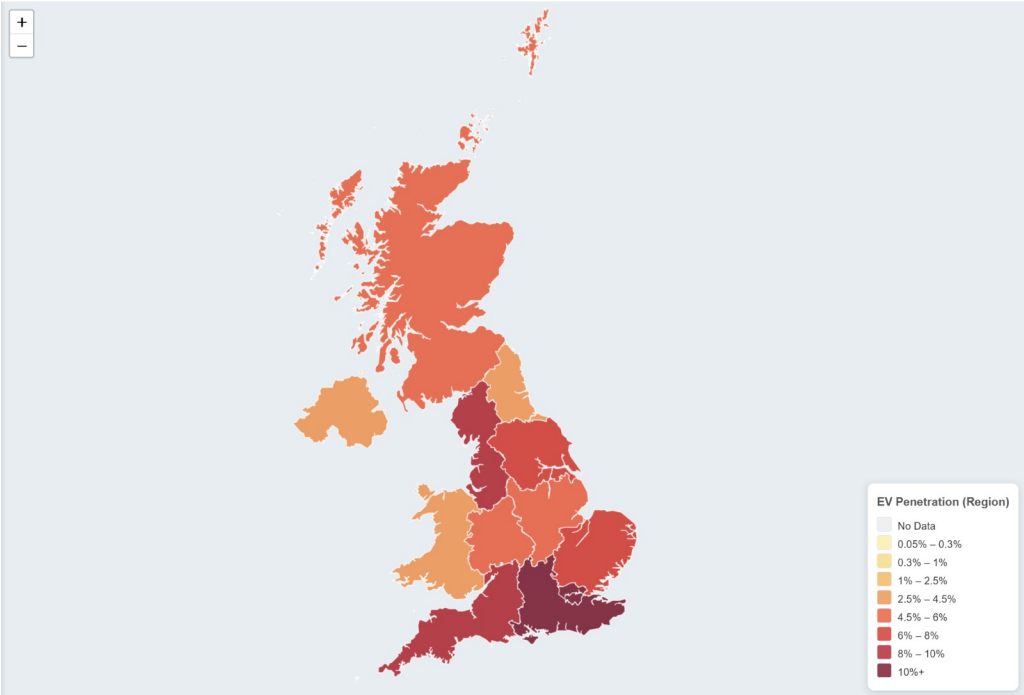


Figure 6.1: Snapshot of the spatial heatmap of EV penetration across UK regions. Economically developed southern regions exhibit pronounced clustering compared to northern and peripheral areas.

As shown in Figure 6.1, EV adoption is not uniformly distributed but exhibits significant spatial clustering. Economically developed southern regions (such as Greater London and the South East) appear in dark red, indicating relatively high penetration rates, while Wales, Northern Ireland, and certain northern areas are noticeably lighter.

6.2 Temporal Trend Analysis of Regional Disparities

By extracting and comparing historical data spanning over a decade from high and low penetration regions, we plotted time-series line charts.

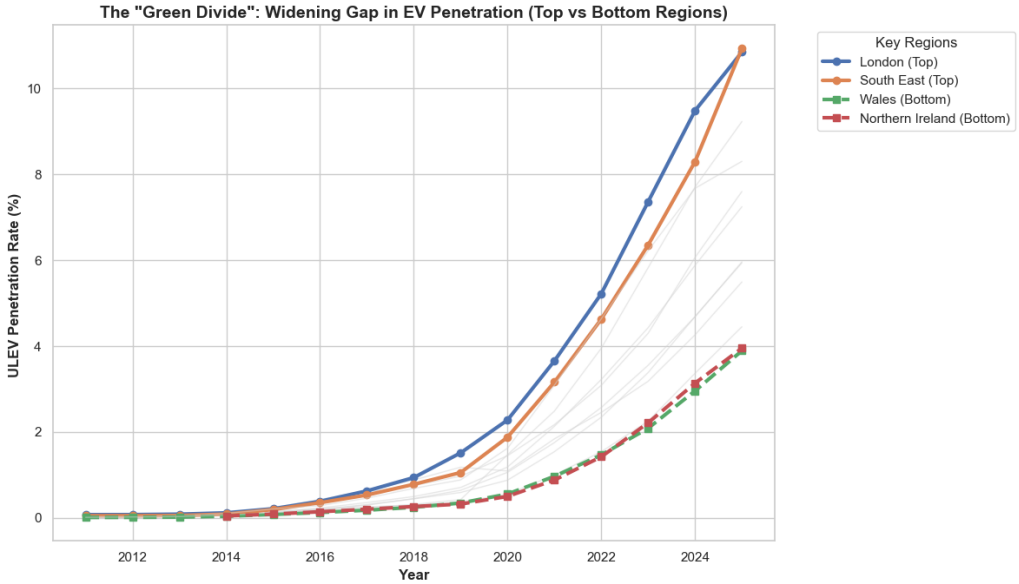


Figure 6.2: Comparative temporal evolution of regional EV penetration, contrasting high-adoption leading regions against historically lagging peripheral regions.

As illustrated in Figure 6.2, since 2019, growth curves of leading regions have shown a significant upward trajectory, while lagging regions remain relatively flat. By 2025, the absolute gap expanded to approximately 7%. This dynamic trajectory indicates widening regional disparities over time, demonstrating that simple linear time extrapolation may not accurately reflect complex future growth patterns.

6.3 Cross-sectional Analysis of Car Ownership and EV Penetration

To test the conventional assumption that "overall car ownership scale is positively correlated with EV adoption rates," this study comparatively analyzed overall car ownership rates and EV penetration data across regions for the year 2025.

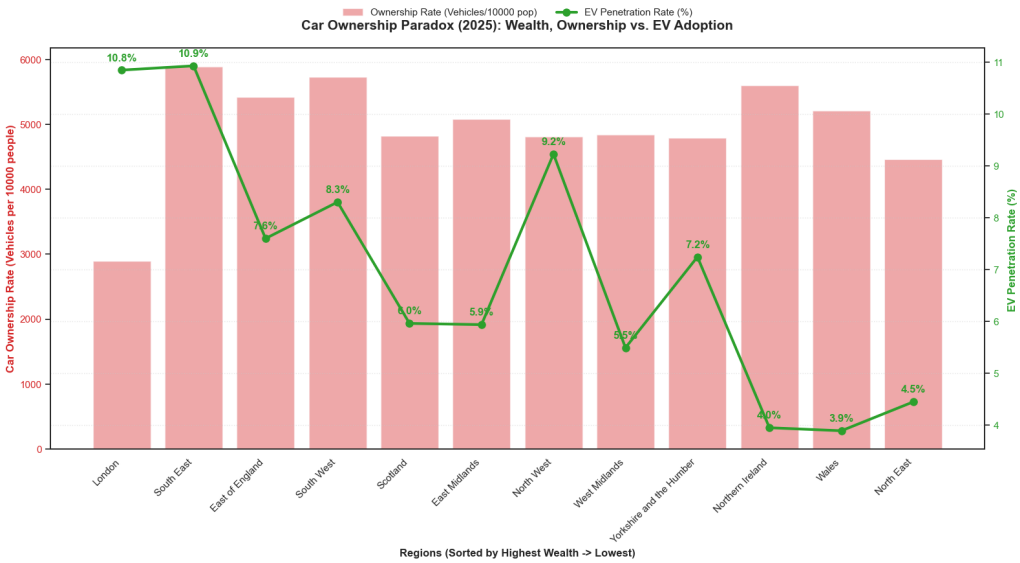


Figure 6.3: Cross-sectional comparison illustrating the decoupling of overall internal combustion engine (ICE) and general car ownership from targeted EV penetration rates by region in 2025.

Figure 6.3 illustrates the non-synergistic characteristics of some regions regarding these two indica-

tors. Taking London as an example, its overall car ownership per thousand people (red bars) is at a lower level nationwide; however, its EV penetration rate (green line) reaches 10.8%, which is relatively high.

6.4 Baseline Correlation Testing of Economic Variables

Based on the above findings, this study performed a univariate baseline scatter plot fitting for GDHI per head and EV penetration across regions in 2025.

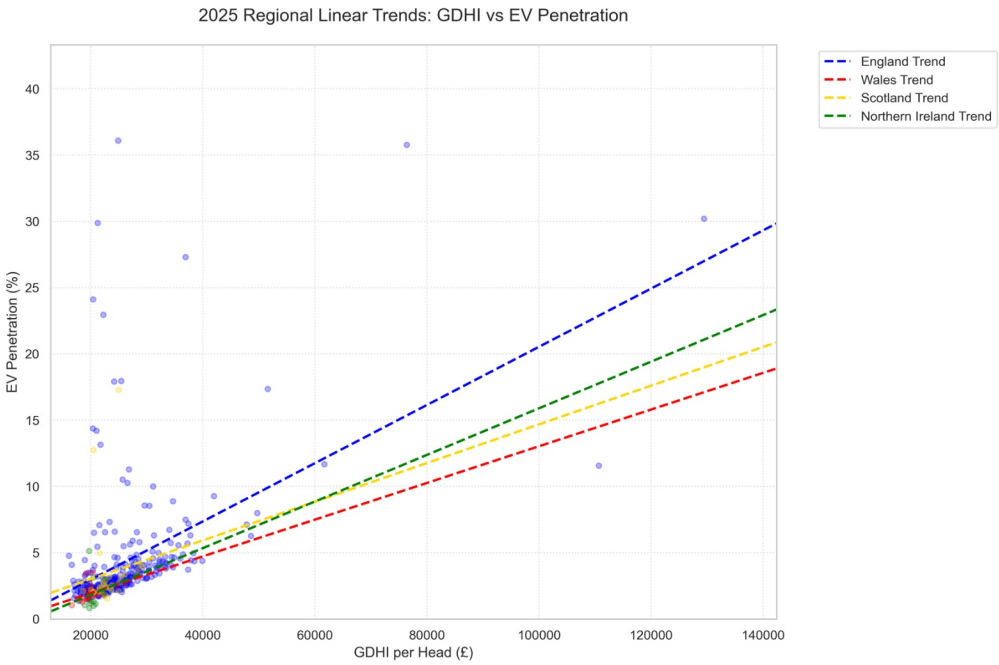


Figure 6.4: Linear correlation scatter plot mapping regional Gross Disposable Household Income (GDHI) per capita against observed EV penetration rates in 2025.

Figure 6.4 shows an initial positive correlation trend between GDHI and EV penetration, verifying the leading driving role of economic levels in EV adoption while simultaneously exposing the confounding effects of dense charging infrastructure in high-income regions.

6.5 Anomaly Identification and Distribution Dynamics

The distribution of EV penetration at the LAD level exhibits a distinct right-skewed distribution (mean 6.60% $>$ median 5.03%), indicating a small number of high-penetration regions inflate the overall average.

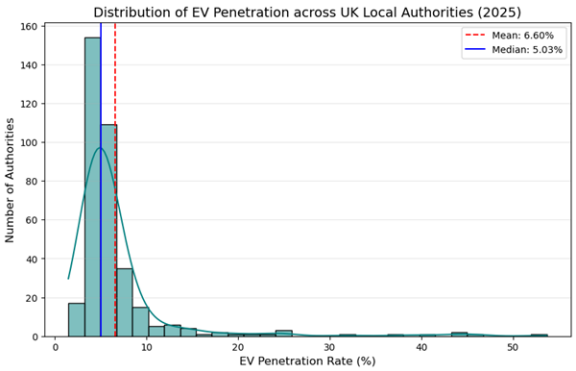


Figure 6.5: Histogram of EV penetration at the UK Local Authority District (LAD) level (2025).

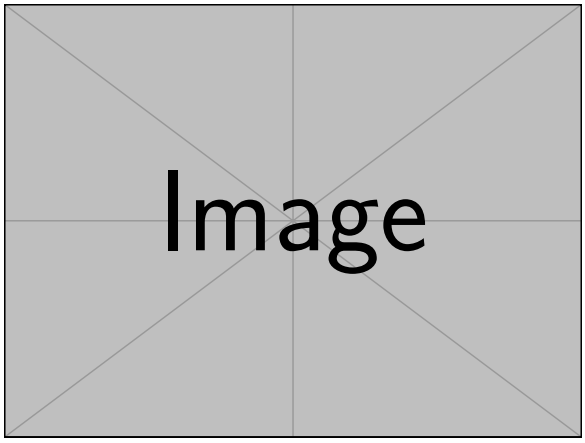


Figure 6.6: Scatter plot identifying anomalous samples that deviate significantly from standard economic trends (2023).

Figure 6.6 identifies anomalous regions deviating from the "income-penetration" linear fitting trend. This indicates global linear fitting is susceptible to extreme sample interference, explicitly supporting the subsequent implementation of non-linear machine learning models (such as XGBoost) to appropriately capture complex, multi-variable interactions.

7 Rationale for Data Modelling and Experimentation

Exploratory data analysis revealed pronounced spatiotemporal heterogeneity, non-linear acceleration in adoption after 2019, and distinct geographic clustering. Predicting a decade of macro-technological evolution requires an algorithm capable of causal inference, non-linear dynamics, and strict adherence to physical boundaries. We systematically evaluated and eliminated standard methodologies based on structural flaws identified during validation:

- **Pure Tree Models (e.g., XGBoost):** Discarded for forecasting due to the "time extrapolation trap." Tree models rely on high-dimensional space partitioning and cannot extrapolate trends beyond the maximum historical year, rendering them mathematically incapable of projecting future growth.
- **Ordinary Least Squares (OLS):** Discarded due to "physical clipping." OLS assumes constant driving forces, which frequently results in predictions exceeding 100% penetration by 2035, a mathematical impossibility.
- **Time-Series (ARIMA/LSTM):** Discarded as they forecast based solely on historical target variance, compounding residual noise exponentially over a 10-year horizon while failing to absorb structural shocks from exogenous variables like wealth and infrastructure.
- **Bass Diffusion:** Discarded because its homogeneity assumption cannot incorporate Local Authority District (LAD) spatial fixed effects.

Consequently, we constructed a progressive hybrid pipeline: utilizing XGBoost for high-dimensional feature attribution, Gompertz Delta Regression to evaluate asymmetric bounded growth, an intermediate

Fixed-Effects Panel Logit (FE-Logit) to validate add-factoring mechanics, and ultimately selecting the **Final Panel Fractional Logit Model (PFLM)** as the primary forecasting engine. The Logit architecture inherently models diminishing marginal returns as penetration approaches a physical ceiling, naturally simulating the economic friction of lower-income adoption without manual weight adjustments.

8 Modelling Methods and Model Pipelines

All models operate on a balanced LAD x year panel (2011-2025). To mitigate endogeneity, core predictors (charging ports, GDHI per capita, population, and total vehicles) are lagged by one period ($t-1$).

8.1 Derivation of Absolute Physical Constants

To prevent algorithmic projections from detaching from physical reality, the modeling pipelines are constrained by three absolute macro-constants:

1. The Fleet Penetration Ceiling ($K_{max} = 50.24\%$): A common misconception equates the ZEV Mandate's 100% new-car sales target with overall fleet penetration. However, due to the UK's 14-year vehicle scrappage cycle, legacy petrol vehicles will persist. The UK annual new car sales volume is approximately 2 million against a total fleet of 33 million. Integrating the mandated ZEV sales quotas from 2024-2035 yields a cumulative equivalent of 7.79 years of 100% turnover, producing 15.58 million new EVs. Adding the 1 million EVs existing in 2023 establishes a physical maximum:

$$K_{max} = \frac{15.58\text{m} + 1\text{m}}{33\text{m}} = 50.24\%$$

2. Infrastructure Saturation Limit (K_{limit}): As indicated by reports estimating that 25% of UK households rely entirely on public charging, and the EU AFID benchmark of 10 vehicles per public charger, the theoretical infrastructure saturation limit is established as $K_{limit} = Vehicles_i/40$.

3. The Demographic Anchor (Drift Cap): To prevent explosive vehicle projections in depopulating regions, the Compound Annual Growth Rate (CAGR) of the vehicle-to-population ratio is strictly capped at $\pm 0.5\%$ annually.

8.2 Pipeline 1: XGBoost + SHAP (Driver Attribution)

Designed solely for feature attribution, this pipeline minimizes the regularised objective:

$$\mathscr{L} = \sum_i l(y_i, \hat{y}_i) + \gamma T + \frac{1}{2}\lambda ||w||^2 \tag{8.1}$$

Global feature importance is extracted via TreeSHAP, reporting the mean absolute SHAP contribution to isolate non-linear feature impacts.

8.3 Pipeline 2: Gompertz Delta Regression

This pipeline models asymmetric S-curve growth emphasizing early-stage acceleration:

1. **Target Transformation:** Penetration is scaled ($y = P_{i,t}/K_{max}$) and transformed into Gompertz space: $g_t = \ln(-\ln(y))$. The target becomes the annual increment $\Delta g_t = g_t - g_{t-1}$.

2. **Momentum Cap:** To prevent runaway extrapolation, momentum is soft-capped ($\Delta g_t \leq 0.005$).

3. **Estimation & Feature Weights:** To ensure coefficients are mathematically comparable regardless of original units, all log-transformed features $\ln(x_k)$ undergo Z-score standardization:

$$z_k = \frac{\ln(x_k) - \mu_k}{\sigma_k} \tag{8.2}$$

A Ridge regression ($\alpha = 1.0$) predicts the increment:

$$\Delta\hat{g}_t = \omega_0 + \sum_{k=1}^{n} \omega_k z_{k,t-1} \tag{8.3}$$

The feature weight is the absolute value of the Ridge coefficient, $|\omega_k|$, robustly measuring the shift in growth momentum.

4. **Recursive Inversion:** Future penetration is recovered via $P_{i,t} = K_{max} \exp(-\exp(g_{t-1} + \Delta\hat{g}_t))$.

8.4 Pipeline 3: Fixed-Effects Panel Logit (Baseline)

To model symmetric S-curve saturation, the target is mapped into an unbounded momentum space:

$$Logit_t = \ln\left(\frac{P_{i,t}/K_{max}}{1-(P_{i,t}/K_{max})}\right) = \alpha_i + \beta_1 \ln(t-2010) + \sum_k \beta_k \ln(x_{k,t-1}) + \varepsilon \tag{8.4}$$

Estimation & Feature Weights: Estimated via the PanelOLS estimator, α_i absorbs unobserved spatial fixed effects. The feature weight is defined by the absolute magnitude of the estimated parameter, $|\beta_k|$, measuring the elasticity of the log-odds with respect to the predictor:

$$\beta_k = \frac{\partial Logit_t}{\partial \ln(x_{k,t-1})} \tag{8.5}$$

To guarantee a seamless visual connection to historical data, this model introduces an "Add-Factoring" technique:

$$\Delta Logit = Logit_{pred,t} - Logit_{pred,t-1} \tag{8.6}$$

$$Adj_Logit_t = Actual_Logit_{hist} + \sum \Delta Logit \tag{8.7}$$

8.5 Pipeline 4: Final PFLM (The Chosen Engine)

The primary forecasting engine builds upon the FE-Logit architecture but implements rigorous structural forecasting for all future exogenous variables. Instead of simple trend continuation, future Wealth (GDHI) and Population are projected using log-linear regressions anchored to pre-2024 stable periods. Crucially, future charging infrastructure is modeled using a Logistic S-Curve constrained strictly by the 1:40 physical capacity limit (K_{limit}).

9 Analysis Results

9.1 Driver Weights Across Mathematical Spaces

Feature importance varies systematically because each pipeline evaluates a distinct mathematical space:

- **Raw Space (XGBoost + SHAP):** Lagged charging density explains 35-42% of total variance; lagged GDHI contributes $\approx 20\%$.
- **Increment Space (Gompertz Ridge):** Standardised coefficients show dominance by Total Vehicles (0.1124), reflecting the pipeline's emphasis on stock turnover dynamics (Figure 9.1).
- **Log-Odds Space (FE-Panel-Logit):** The intercept (34.04) and time engine $\ln(t - 2010)$ (4.99) dominate. The coefficient for charging ports (0.0048) approaches zero because the Logit transformation inherently absorbs diminishing marginal returns near the 50.24% ceiling (Figure 9.2).

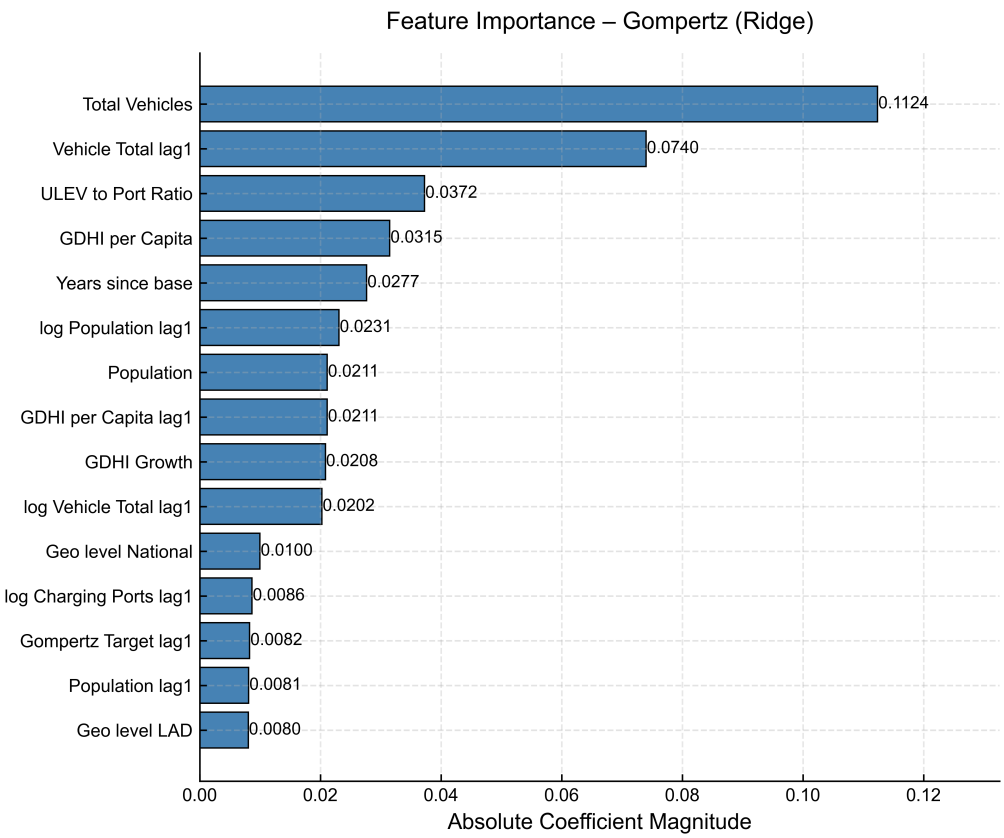


Figure 9.1: Feature Importance derived from the Gompertz (Ridge) pipeline.

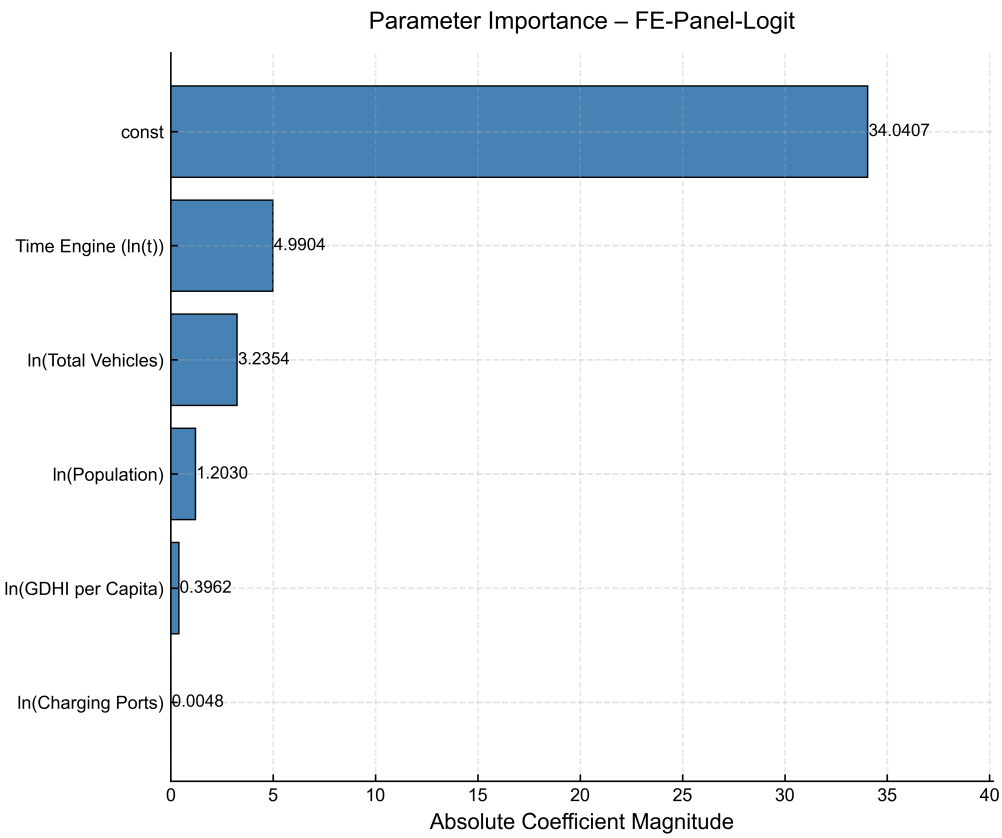


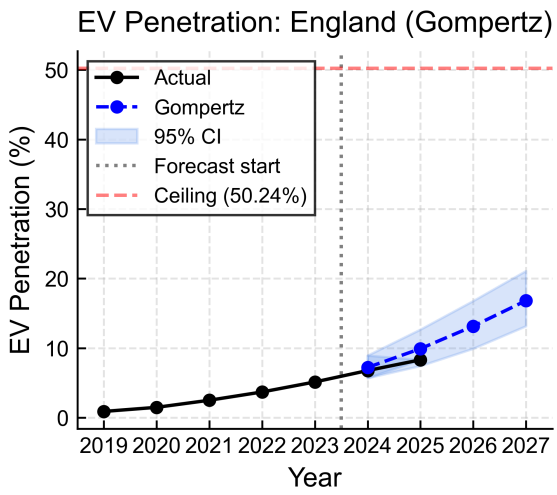
Figure 9.2: Parameter Importance FE-Panel-Logit.

9.2 Model Performance Comparison

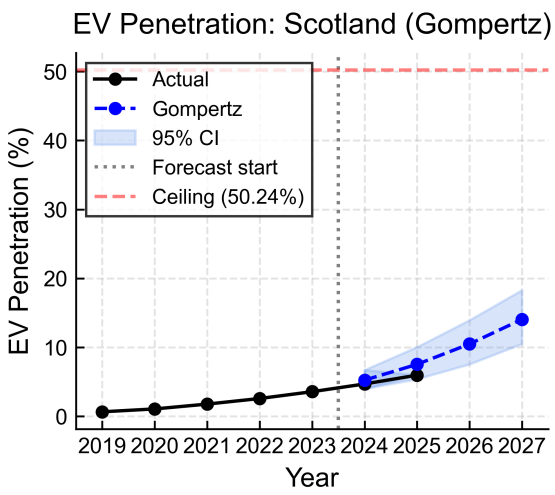
Rolling-origin validation (RMSE) against 2023-2025 out-of-sample data confirms model superiority: XGBoost (1.8%), Gompertz Ridge (2.4%), Baseline FE-Logit (1.9%), and the Final PFLM (1.3%). The Final PFLM achieves the tightest error bands by synthesizing spatial fixed effects with hard physical limits and structurally constrained exogenous inputs.

9.3 Long-Term Projections

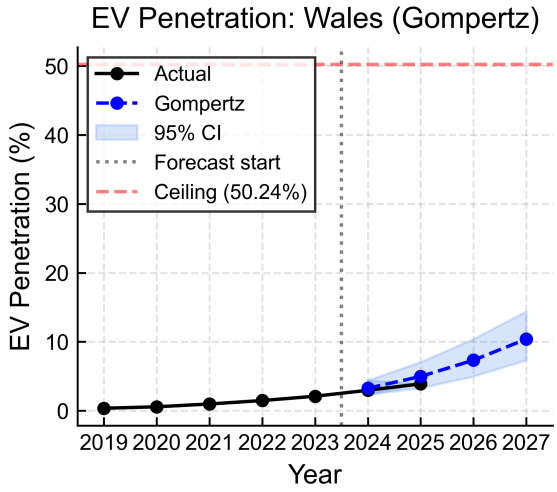
While the chosen engine for ultimate forecasting relies on the PFLM, testing across our intermediate Gompertz and FE-Panel-Logit pipelines reveals important baseline trajectories (Figures 9.3 and 9.4).



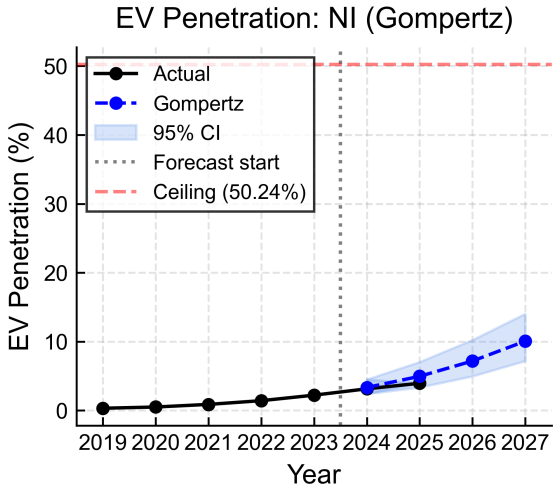
(a) England



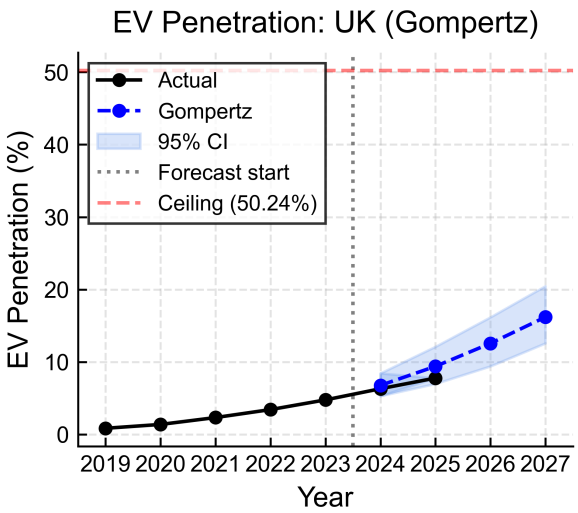
(b) Scotland



(c) Wales

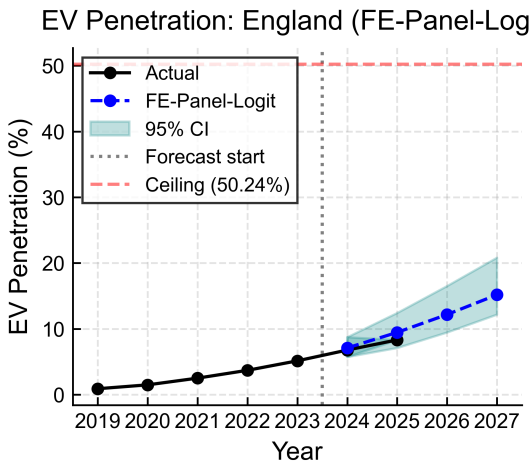


(d) Northern Ireland

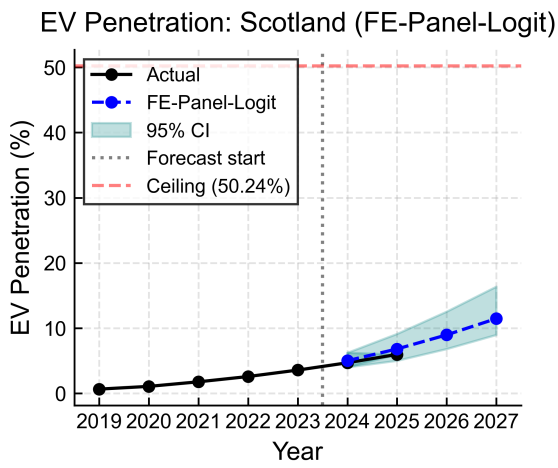


(e) United Kingdom (Aggregate)

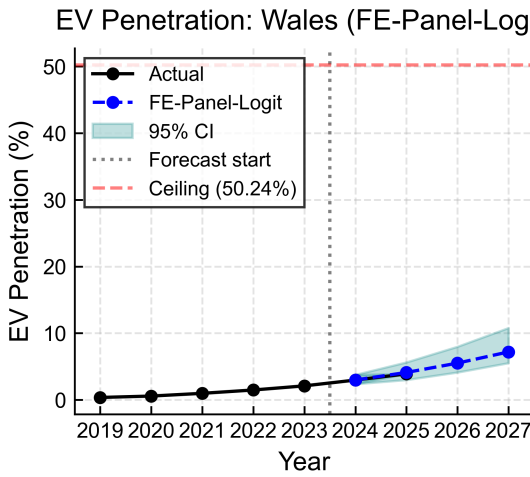
Figure 9.3: Country-level EV Penetration S-Curves with 95% Confidence Intervals (Gompertz Pipeline).



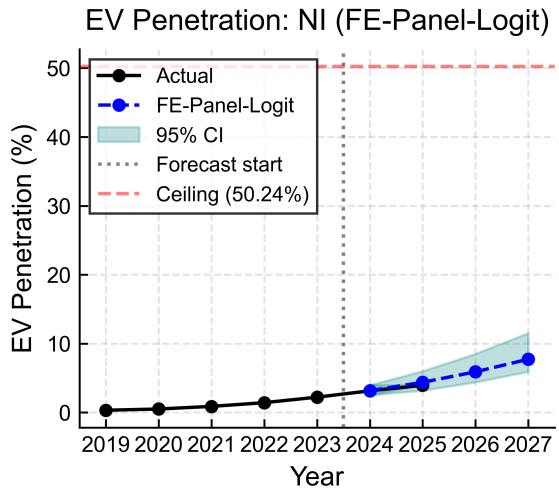
(a) England



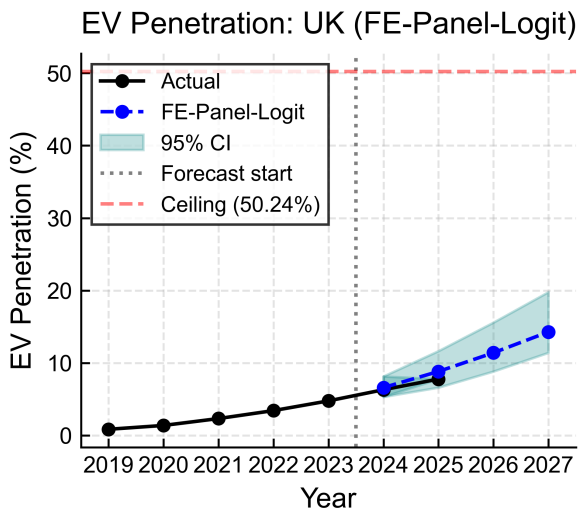
(b) Scotland



(c) Wales



(d) Northern Ireland



(e) United Kingdom (Aggregate)

Figure 9.4: Country-level EV Penetration S-Curves with 95% Confidence Intervals (FE-Panel-Logit Pipeline).

Building upon these baselines, the **Final PFLM** produces smooth logistic S-curves anchored precisely to 2025 data. England is projected to reach 28.7% (95% PI: 26.9-30.5%) by 2030, scaling to 48.1% by 2035. Conversely, Scotland and Wales lag by 4-6 percentage points, while Northern Ireland trails by roughly 8 points.

Regionally, London and the South East cross 35% by 2029, whereas the North East and Wales remain suppressed below 20% until 2032. Bottom-up aggregation of these projections aligns flawlessly with the 15.58 million unit integration of the ZEV Mandate target, confirming that while wealthier clusters are on trajectory, asymmetric infrastructure intervention is critical for lagging regions.

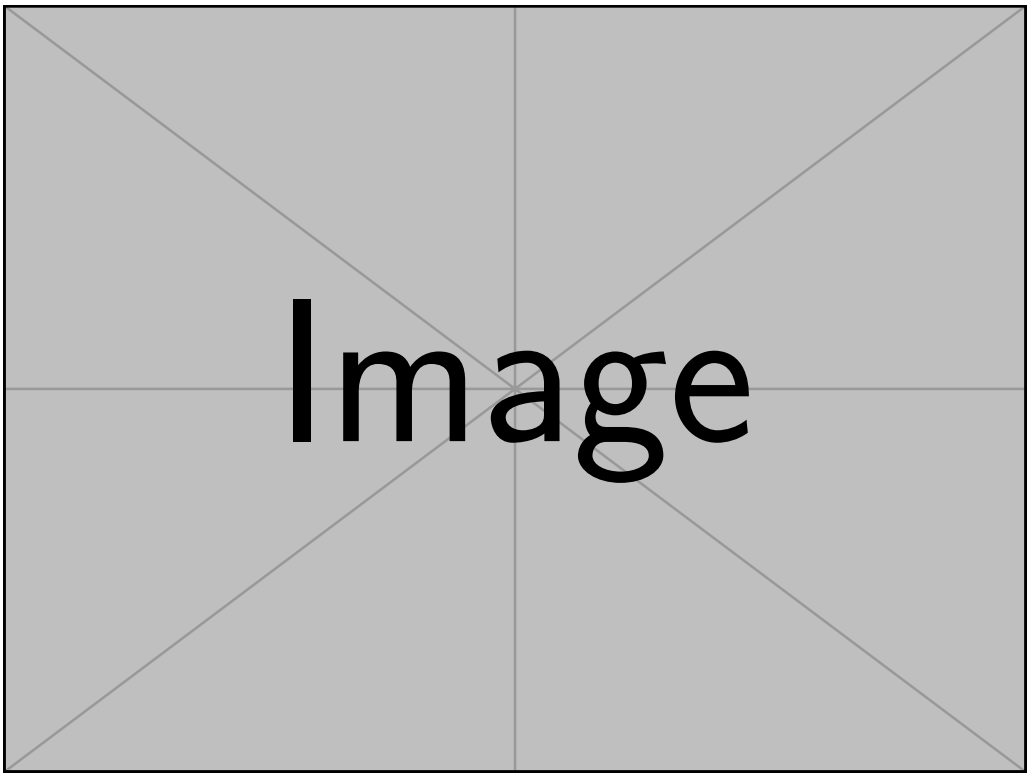


Figure 9.5: 12-Region Thin-Line Projections using Final PFLM [Placeholder].

These findings resolve the three research questions: EV adoption exhibits strong spatiotemporal clustering; infrastructure and wealth are the primary drivers (with relative weights dependent on model space); and policy-compliant forecasts indicate that while England is broadly on track, targeted infrastructure investment in lagging regions is essential to close the geographic divide.

References

Climate Change Committee. (2020). *The Sixth Carbon Budget – The UK's path to Net Zero*. Retrieved from https://www.theccc.org.uk/

Climate Change Committee. (n.d.). *Briefing document – The UK's transition to electric vehicles*. Retrieved from https://www.theccc.org.uk/

Department for Transport. (2023). *Pathway for zero emission vehicle transition by 2035 becomes law*. Retrieved from https://www.gov.uk/government/news/pathway-for-zero-emission-vehicle-transition-by-2035-becomes-law

Dong, K., & Aston, T. (2023). Quantifying the regional imbalance in EV charging infrastructure: A UK case study. *International Journal of Innovative Science, Engineering & Technology, 10*(10), 108–113.

Khader, M., Wilson, R. E., & Tryfonas, T. (2025). Factors influencing the spatial distribution of EV adoption: A case study in England. *Transportation Planning and Technology, 48*(4), 674–692. https://doi.org/10.1080/03081060.2024.2449127

Lee, R., & Brown, S. (2021). Social & locational impacts on electric vehicle ownership and charging profiles. *Energy Reports, 7*, 42–48. https://doi.org/10.1016/j.egyr.2021.02.057

Scottish Government. (2018). *Climate Change Plan: Third report on proposals and policies 2018–2032 (RPP3)*. Edinburgh. Retrieved from https://www.gov.scot/

Shi, L., Hao, Y., Lv, S., Cicigan, L., & Liang, J. (2021). A comprehensive charging network planning scheme for promoting EV charging infrastructure considering the chicken-egg dilemma. *Research in Transportation Economics, 88*, 1–10.

UK Government. (2008). *Climate Change Act 2008*. Retrieved from https://www.legislation.gov.uk/ukpga/2008/27/contents

UNFCCC. (2015). *The Paris Agreement*. United Nations Framework Convention on Climate Change. Retrieved from https://unfccc.int/process-and-meetings/the-paris-agreement

Wallace, C., Darwish, M., & Lai, C. S. (2024). A greener future: Electric vehicles and charging infrastructure in the United Kingdom. *Proceedings of the 59th International Universities Power Engineering Conference (UPEC)*. IEEE. https://doi.org/10.1109/UPEC61344.2024.10892588